Chapter 1. Introduction to Nonparametric Statistics

Nonparametric models, parametric inference review, why nonparametric methods, robustness

Contents

1.1 What is nonparametric statistics? 2

1.1.1 Nonparametric models . . . 2

1.1.2 Nonparametric statistics and parameters . . . 2

1.1.3 Three tools of nonparametric statistics . . . 3

1.2 Parametric models and statistical inference 4

1.2.1 Standard parametric models . . . 4

1.2.2 Parameter estimation and hypothesis testing . . . 5

1.3 Why nonparametric methods are needed 8

1.4 Distribution-free and robust methods 8

1.1 What is nonparametric statistics?

Nonparametric statistics (nonparametric statistical methodology) is the theory and practice of statistical inference (estimation and hypothesis testing) under *few assumptions about the population.* Three guiding questions:

1. To compare the means of two samples we would use a t-test if both were normal. Which test should we use if normality is badly violated?
2. The coefficient of variation (standard deviation divided by the mean) is a common index of inequality. From the scholarship amounts $x_1, \ldots, x_n$ of $n = 100$ randomly chosen students we compute $s/\bar{x}_n$. How accurate is this estimate?
3. Poll results for a politician's approval are observed over time (Figure 1.1 of the book). The relation between time and approval is clearly not linear. How can we estimate the change of approval over time accurately?

The first question (testing without normality) is answered by classical nonparametric, *distribution-free* methods: tests based on signs and ranks, and permutation tests. The second concerns the uncertainty of an estimator; the mathematical-statistical calculation is hard in general, and the *bootstrap* replaces it by computation. The third concerns estimating a curve; *nonparametric function estimation* (smoothing) uses the observations near each point to estimate the curve there.

1.1.1 Nonparametric models

Assumptions about a population are usually made through a parametric model. If $X_1, \ldots, X_n$ are drawn from a population, one often assumes that its density f belongs to the **normal family**

$$\begin{gathered} \mathcal{F}_G = \{\phi_{\mu,\sigma} : \mu \in (-\infty, \infty),\ \sigma > 0\}, \\ \phi_{\mu,\sigma}(x) = \phi(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(- \frac{(x-\mu)^2}{2\sigma^2} \Big), \quad -\infty < x < \infty. \end{gathered} \tag{1.1}$$

This is the normality assumption, a *parametric model*: the population is determined by the two parameters μ, σ. (A "parameter" is mathematically an undetermined constant, and in computing a variable that determines a property of a system.) Nonparametric statistics avoids such a model and offers methods valid whatever family f belongs to. Examples of families used instead:

- $\mathcal{F}_1 = \{f : \int x^2 f(x)\, dx < \infty\}$, all distributions with finite mean and variance;
- $\mathcal{F}_2 = \{f : f(\mu - x) = f(\mu + x),\ x \in (-\infty, \infty) \text{ for some } \mu\}$, distributions whose density is symmetric about some μ (the mean need not exist).

These families contain uncountably many distributions, and parametrizing them needs infinitely many parameters, whereas $\mathcal{F}_G$ is indexed by a two-dimensional parameter space. Note $\mathcal{F}_G \subset \mathcal{F}_1$ and $\mathcal{F}_G \subset \mathcal{F}_2$. Most parametric models (normal, gamma, ...) lie in $\mathcal{F}_1$. A strong assumption on the population means a small family; assuming only $f \in \mathcal{F}_1$ is a very weak assumption.

1.1.2 Nonparametric statistics and parameters

In nonparametric statistics a **parameter** is a characteristic of an element of the family. For $f \in \mathcal{F}_1$, the mean $\mu_f := \int x f(x)\, dx$ and variance $\sigma_f^2 = \int (x - \mu_f)^2 f(x)\, dx$ are parameters; but knowing them does not determine f. For $f \in \mathcal{F}_1$ or $\mathcal{F}_2$ the median is also a parameter. Nonparametric inference is estimation and testing of parameters defined on a very large family; when the assumptions are extremely weak (e.g. no normality, only $f \in \mathcal{F}_2$), the methods are called **distribution-free**.

A parameter may be simple (mean, median) or anything that characterizes the population. Under normality all quantiles are determined by the mean and variance; under $\mathcal{F}_1$ they are not, and a quantile (percentile) is itself a parameter to estimate. The density f or the distribution function $F(x) = \int_{-\infty}^{x} f(t)\,dt$ may also be the target. Accordingly:

1. **Location and scale parameters**: mean, median, variance. Classical distribution-free methods transform the data into signs, ranks or scores; e.g. the Wilcoxon signed-rank test, the Kruskal–Wallis test, the Friedman test.
2. **The density as a parameter**: the field of *density estimation*.
3. **Parameters in complex models**: extensions to ANOVA, regression, multivariate analysis.

For instance, the usual simple linear regression model assumes

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

With the X_i fixed, the Y_i are independent normal with means $\beta_0 + \beta_1 X_i$ and variance σ^2, so the family

$$f(y_1, \ldots, y_n) \in \Big\{ \prod_{i=1}^{n} \phi(y_i; \beta_0 + \beta_1 X_i, \sigma^2) : \beta_0, \beta_1 \in \mathbb{R},\ \sigma^2 > 0 \Big\}$$

is a parametric model with three parameters (ϕ as in (1.1)). The *distribution-free* linear regression model is

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} f, \quad f \in \mathcal{F}_1,$$

and a *nonparametric* regression model is, e.g.,

$$Y_i = m(X_i) + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2),$$

where $m(\cdot)$ is any curve in a function space $\mathcal{H}$ (e.g. smooth or differentiable functions). Both $\mathcal{F}_1$ and $\mathcal{H}$ are infinite-dimensional, so these families cannot be described by finitely many parameters. In distribution-free regression the interest is in β_0, β_1, approached much like the median in a one-sample model; in nonparametric regression the function-valued parameter $m(\cdot)$ is estimated.

1.1.3 Three tools of nonparametric statistics

Nonparametric methods are needed because:

- they are developed under very weak assumptions (or very large model spaces), so errors from violating assumptions are small;
- they also apply under parametric models, and mathematical calculation can be replaced by computation;
- the model space is so large that they perform very well in estimating complex systems.

Correspondingly there are three kinds of methodology:

1. **Permutation tests.** Proposed by R. A. Fisher; widely used since the 1990s as computation became cheap. They follow Fisher's idea of *conditional inference*: inference given the observed data, i.e. treating the observed values as fixed.
2. **The bootstrap** (Bradley Efron). Obtains the sampling distribution of a statistic from the given sample, without a parametric model. Like permutation tests it relies on computation and repetition; both are *resampling* methods.

3. **Smoothing.** Estimating a density $f(\cdot)$ or regression function $m(\cdot)$ without a specific form is hard because the function space is infinite-dimensional. Smoothing estimates $m(x)$ or $f(x)$ at a point x from the observations near x; as x varies, the observations used change smoothly, giving a smooth estimate $\hat{m}(\cdot)$.

Classical sign and rank methods were mostly developed before the 1970s and can be understood as special permutation tests. From the 1980s, smoothing-based nonparametric function estimation (density estimation, nonparametric regression) became the mainstream, while permutation tests and the bootstrap proved highly effective for complex statistics of complex populations, under both parametric and nonparametric models.

1.2 Parametric models and statistical inference

Statistical inference splits into estimation and hypothesis testing. To explain point and interval estimation, standard errors and testing, we first review standard parametric models.

1.2.1 Standard parametric models

Normal distribution. If $X_1, \ldots, X_n$ come from a normal population, the family is $\mathcal{F}_G$; we write $X \sim N(\mu, \sigma^2)$. The density (1.1) is determined by the location parameter $\mu \in (-\infty, \infty)$ and the scale parameter $\sigma^2 > 0$, with $E(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$. $N(0,1)$ is the *standard normal*, and any $X \sim N(\mu, \sigma^2)$ standardizes to $Z = (X - \mu)/\sigma \sim N(0,1)$. With $\Phi(z) = F_Z(z) = P(Z \le z)$,

$$F_X(x) = P(X \le x) = P\{Z = (X-\mu)/\sigma \le (x-\mu)/\sigma\} = \Phi\{(x-\mu)/\sigma\}.$$

Definition (Quantile)

For a distribution with distribution function F and $p \in (0,1)$, the **p-quantile** is

$$Q_p = \min\{z : F(z) \ge p\},$$

the smallest point at which the cumulative probability reaches p. As a function of p it is the **quantile function** $Q(p) = Q_p$.

If F is invertible, $Q_p = F^{-1}(p)$; for the normal, the quantile function is $\Phi^{-1} : (0,1) \to (-\infty, \infty)$. For a significance level $\alpha \in (0,1)$ the **z-value** is $z_\alpha = \Phi^{-1}(1-\alpha)$; e.g. $z_{0.05} = \Phi^{-1}(0.95) \approx 1.645$, $z_{0.025} = \Phi^{-1}(0.975) \approx 1.96$.

Binomial distribution. A Bernoulli trial has two outcomes (success/failure, positive/negative, ...), coded 1 and 0. If independent Bernoulli trials have common success probability $p = P(\text{“1”}) \in (0,1)$, the number X of successes in n trials follows the binomial distribution $X \sim B(n,p)$; the parameter is p (n is fixed). Its probability mass function and distribution function are

$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n,$$

$$F_B(x) = P(X \le x) = \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k (1-p)^{n-k}.$$

F_B is a step function, so it is not invertible, but $Q_p = \min\{x : F_B(x) \ge p\}$ is still defined, and:

- for an integer $x \in \{0, \ldots, n\}$, if $p \in (F(x-1), F(x)]$ then $Q_p = x$;

- for every $p \in (0,1)$, $F(Q_p - 1) < p \le F(Q_p)$, hence

$$P(X \le Q_p) \ge p, \qquad P(X \ge Q_p) > 1 - p.$$

Proof (supplied). F is nondecreasing, right-continuous and constant on each $[x, x+1)$, so $\{z : F(z) \ge p\} = [x_0, \infty)$ for the smallest integer x_0 with $F(x_0) \ge p$, and $Q_p = x_0$. Minimality gives $F(x_0 - 1) < p \le F(x_0)$. If $p \in (F(x-1), F(x)]$ then x is that smallest integer, so $Q_p = x$. Finally $P(X \le Q_p) = F(Q_p) \ge p$ and $P(X \ge Q_p) = 1 - F(Q_p - 1) > 1 - p$. ■

The mean and variance are $E(X) = \sum_{x=0}^{n} x f(x) = np$ and $\mathrm{Var}(X) = np(1-p)$ (X is a sum of n independent Bernoulli(p) variables, each with mean p and variance $p(1-p)$).

1.2.2 Parameter estimation and hypothesis testing

Example 1.1

Eight randomly chosen men in their 50s joined a weight-loss program. After 10 weeks their weight losses (kg) were $3, 2, 0, 8, 12, 11, 5, -2$ (the last participant gained weight). How much weight loss can a man in his 50s expect from the program, and is the program effective?

The first question is estimation, the second hypothesis testing.

Estimation. The population in Example 1.1 is the distribution of weight losses, and the "expected loss" is a location parameter (mean, median, mode). Under normality ($f \in \mathcal{F}_G$) mean, median and mode all equal μ, and $\bar{X}_n = \frac{1}{n}\sum X_i$ is known to be the best estimator. In particular:

1. $E(\bar{X}_n) = \mu$: it is **unbiased**, with **standard error** $\mathrm{se} = (\mathrm{Var}(\bar{X}_n))^{1/2} = \sigma/\sqrt{n}$, the smallest among unbiased estimators.
2. $\bar{X}_n \sim N(\mu, \sigma^2/n)$ exactly; if σ^2 is known an exact confidence interval follows: the 95% interval $\mu \in \bar{X}_n \pm 1.96\sigma/\sqrt{n}$ has exact coverage because $Z = (\bar{X}_n - \mu)/\mathrm{se} \sim N(0,1)$.
3. If σ^2 is unknown, studentize with $s^2 = \sum_{i=1}^{n}(X_i - \bar{X}_n)^2/(n-1)$:

$$T = \frac{\bar{X}_n - \mu}{\widehat{\mathrm{se}}}, \qquad \widehat{\mathrm{se}} = \frac{s}{\sqrt{n}}.$$

   Under normality $T \sim t_{n-1}$, and the 95% interval is $\mu \in \bar{X}_n \pm t_{0.025,n-1}s/\sqrt{n}$, where $t_{\alpha,n-1}$ is the $1-\alpha$ quantile of t_{n-1}.

Unbiasedness and small standard error are desirable. Under a parametric model (e.g. normal) one can find an optimal estimator that is unbiased with the smallest standard error; mathematical statistics is largely about finding optimal estimators under various parametric assumptions. But if the population is far from normal, the standard error of the estimator (the measure of accuracy) can be very large or very small compared with $\widehat{\mathrm{se}}$, and the standard error of $\bar{X}_n$, "optimal" under normality, can be much larger than that of other estimators. The nonparametric methods to come guarantee a certain performance for any population in a very large family, i.e. under very weak assumptions.

Hypothesis testing. The second question of Example 1.1 is a test: choose between two hypotheses about a parameter, the null H_0 and the alternative H_1, using the data. "Is the program effective?" becomes $H_0 : \mu = 0$ (no effect) versus $H_1 : \mu > 0$ (effect). A **test statistic** takes typically different values under the two hypotheses; e.g. $T = \bar{X}_n/\widehat{\mathrm{se}}$ tends to be large under H_1 and near 0 under H_0 (where $T \sim t_{n-1}$). The decision rule usually has the form

- if $T > c$, reject H_0 (accept H_1);
- if $T < c$, fail to reject H_0 (insufficient evidence against H_0).

The **rejection region** determined by c is chosen so that the probability of a type I error (rejecting a true H_0) is at most the significance level α (usually 0.05). The closer the type I error probability is to α, the higher the **power** (probability of accepting H_1 when it is true), so if possible c is set so that the type I error probability is exactly α. For the t-test the **critical value** is $c = t_{\alpha,n-1}$: reject when $T > t_{\alpha,n-1}$. Under normality the t-test is the most powerful among tests of the location parameter with type I error controlled at α. If normality fails badly, the t-test no longer guarantees control of the type I error at the given level; permutation tests, in particular, control it exactly under very weak assumptions.

- Tests are classified by the direction of H_1. For $H_0 : \mu = \mu_0$ and $T = (\bar{X}_n - \mu_0)/\widehat{\text{se}}$:
  - lower-tailed: $H_1 : \mu < \mu_0$, rejection region $\{T : T < -t_{\alpha,n-1}\}$;
  - upper-tailed: $H_1 : \mu > \mu_0$, rejection region $\{T : T > t_{\alpha,n-1}\}$;
  - two-sided: $H_1 : \mu \neq \mu_0$, rejection region $\{T : |T| > t_{\alpha/2,n-1}\}$.
- The p-**value** measures how inconsistent the data are with H_0: the probability, under H_0, of a test statistic at least as extreme as the observed T_{obs}, with $T \sim t_{n-1}$:
  - lower-tailed: p-value $= P(T < T_{\text{obs}})$;
  - upper-tailed: p-value $= P(T > T_{\text{obs}})$;
  - two-sided: p-value $= P(T < -|T_{\text{obs}}|$ or $T > |T_{\text{obs}}|)$.

  The smaller the p-value, the harder it is to believe H_0. Rejecting H_0 when the p-value is below α is the same rule as the rejection region above.

The statistician's job in testing: set appropriate assumptions on the population; express the hypotheses mathematically in terms of parameters; choose a test statistic distinguishing H_0 from H_1; find the rejection region. In nonparametric statistics, where the assumptions are very weak, the key is a **distribution-free test statistic**, one that does not depend on the population distribution. Finding the rejection region then means finding the distribution of the statistic under H_0, the **null distribution** (t_{n-1} for the t-statistic under normality).

Example 1.2

Assume the data of Example 1.1 come from a normal population: $X_1, \ldots, X_8 \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$. Answer the questions at $\alpha = 0.05$. The sample mean and standard deviation are $\hat{\mu} = \bar{x}_n = 4.875$ and $\hat{\sigma} = s = 5.1$, so the expected loss is estimated as 4.875 kg, and the 95% confidence interval $\mu \in \bar{x}_n \pm t_{0.025,7} s/\sqrt{8}$ is computed in R:

```
> x = c(3, 2, 0, 8, 12, 11, 5, -2)
> n = length(x)
> (ci = mean(x) + c(-1,1) * qt(1-0.025,n-1) * sd(x) / sqrt(n))
[1] 0.6253086 9.1246914
```

To test effectiveness, take $H_0 : \mu = 0$ versus $H_1 : \mu > 0$ and run the t-test:

```
> t.test(x, alternative = "greater")$p.value
[1] 0.01504372
```

The p-value is below 5%, so the program has a significant effect at $\alpha = 5\%$.

Duality of tests and confidence intervals. Suppose inference on μ uses a point estimate (or its standardized form T) and its sampling distribution. When testing $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$: if μ_0 lies in the 95% confidence interval (L, U), the level-5% test does not reject H_0; if $\mu_0 \notin (L, U)$ it rejects. Conversely, if the level-α test of $H_0 : \mu = \mu'$ fails to reject, μ' lies in the interval. Thus tests can be obtained from confidence intervals and vice versa.

Example 1.3 (Tests and confidence intervals under normality)

For $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, the 95% interval for μ is $CI(\mu) = \bar{X}_n \pm t_{0.025,n-1} s/\sqrt{n}$. The level-5% t-test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$ rejects when $|\bar{X}_n - \mu_0| > t_{0.025,n-1} s/\sqrt{n}$. If the test fails to reject,

$$|\bar{X}_n - \mu_0| \leq t_{0.025,n-1} s/\sqrt{n} \iff \mu_0 \in \bar{X}_n \pm t_{0.025,n-1} s/\sqrt{n}, \tag{1.2}$$

so $\mu_0 \in CI(\mu)$; if it rejects, $\mu_0 \notin CI(\mu)$. Conversely, the test that rejects H_0 exactly when μ_0 is not in the 95% interval is the t-test.

Normality and the central limit theorem. For real data, normality almost never holds exactly. A population with a symmetric, bell-shaped density is approximately normal, and the "normality assumption" is generally accepted in that approximate sense; the parametric methods above (intervals, t-tests) are then approximately correct. Data often differ greatly from normal, e.g. discrete data (from a binomial population), ordinal data (survey answers: very satisfied, satisfied, neutral, dissatisfied, very dissatisfied), or continuous data from a heavily skewed distribution. Still, t-based inference is often used for them, because by the *central limit theorem* the large-sample distribution of the statistic agrees with its null distribution under normality.

Theorem 1.1 (Central limit theorem)

Let $f \in \mathcal{F}_1$ and $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} f$. The sample mean $\bar{X}_n$ has mean $E(\bar{X}_n) = E(X_1) = \mu_f$ and variance $\text{Var}(\bar{X}_n) = \frac{1}{n}\text{Var}(X_1) = \frac{\sigma_f^2}{n}$. When n is large, the standardized sample mean is approximately standard normal:

$$\frac{\bar{X}_n - \mu_f}{\sqrt{\sigma_f^2/n}} \xrightarrow{d} N(0,1),$$

or $\bar{X}_n \dot{\sim} N(\mu_f, \sigma_f^2/n)$.

Proof (supplied). The mean and variance follow from linearity and independence: $E\bar{X}_n = \frac{1}{n}\sum EX_i = \mu_f$ and $\text{Var}\,\bar{X}_n = \frac{1}{n^2}\sum \text{Var}\,X_i = \sigma_f^2/n$. Assume $\sigma_f^2 > 0$ and let $Y_i = (X_i - \mu_f)/\sigma_f$, so $EY_i = 0$, $EY_i^2 = 1$, and $Z_n := \frac{\bar{X}_n - \mu_f}{\sigma_f/\sqrt{n}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n Y_i$. Let $\varphi(t) = Ee^{itY_1}$. Since $E|Y_1|^2 < \infty$, φ is twice differentiable with $\varphi(0) = 1$, $\varphi'(0) = iEY_1 = 0$, $\varphi''(0) = -EY_1^2 = -1$, so $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$ as $t \to 0$. By independence,

$$Ee^{itZ_n} = \varphi(t/\sqrt{n})^n = \left(1 - \frac{t^2}{2n} + o(\tfrac{1}{n})\right)^n \longrightarrow e^{-t^2/2}$$

for every fixed t, using $(1 + a_n/n)^n \to e^a$ when $a_n \to a$ (valid for complex a_n). The limit is the characteristic function of $N(0,1)$, so by Lévy's continuity theorem $Z_n \xrightarrow{d} N(0,1)$. ∎

In Theorem 1.1, $X_n \xrightarrow{d} f$ means that the distribution of X_n converges to f as $n \to \infty$ (*convergence in distribution*); $\bar{X}_n \dot{\sim} f_n$ means that for large n the distribution of $\bar{X}_n$ is approximately f_n. By the CLT, statistics of the form of a mean or sum can be approximated by a normal distribution. Thus

the usefulness of the normal model extends beyond normal-like populations to every population for which the large-sample CLT approximation applies to the estimator or test statistic.

1.3 Why nonparametric methods are needed

Methods developed under parametric (especially normal) models remain useful when the population is not exactly normal. Still, nonparametric methods are needed for three reasons:

- They are developed under very weak assumptions (or very large model spaces), so errors from violated assumptions are very small. In particular, in small samples the normal approximation via the CLT cannot be used.
- They also apply under parametric models, and mathematical calculation is easily replaced by computation. Even with large samples, when the CLT does not apply, the sampling distribution of an estimator or the null distribution of a test statistic can be approximated nonparametrically.
- The nonparametric model space is so large that they excel at estimating complex systems. Just as the normal family approximately contains many similar distributions, nonparametric statistics approximates an infinite-dimensional (nonparametric) family by finite-dimensional families of very high dimension.

1.4 Distribution-free and robust methods

A **robust** statistical method performs well even when the data contain **outliers**.

Example 1.4

The copper content (parts per million) of 24 wholemeal flour products, in increasing order (Analytical Methods Committee, 1989):

```
2.20 2.20 2.40 2.40 2.50 2.70 2.80 2.90
3.03 3.03 3.10 3.37 3.40 3.40 3.40 3.50
3.60 3.70 3.70 3.70 3.70 3.77 5.28 28.50
```

The last value 28.50 looks like an outlier. Whether a value is an outlier should not be decided by the statistician alone but with the judgment of experts of the field: 28.50 may be a typo for 2.85, a measurement-device error, or a genuine value. The mean and standard deviation change drastically with and without it:

- mean: 4.26 (with), 3.21 (without);
- standard deviation: 5.2 (with), 0.68 (without).

If 28.50 really is an outlier, the sample mean and standard deviation should not be used to summarize the data. By contrast, the robust summaries, the sample **median** and the **median absolute deviation** (MAD)[1], give

- median: 3.385 (with), 3.37 (without);
- MAD: 0.53 (with), 0.50 (without),

[1]The MAD measures spread: it is the median of the absolute median-centered deviations $|x_i - \text{median}\{x_i\}|$, $\text{MAD}(x) = \text{median}\{|x_i - \text{median}\{x_i\}|\}$. Compare the standard deviation, the root mean square of the mean-centered deviations, $\text{sd}(x) = \left(\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2\right)^{1/2}$.

and change little. Distribution-free nonparametric methods keep their properties whatever the population. Data with an outlier can be viewed as a sample from a very heavy-tailed population in which very large values occur; so distribution-free methods are robust. (Not every robust method is nonparametric.) Several nonparametric methods in this book are better known as robust methods, and are motivated that way when they appear.